

PACER: A Perturbation-Aware Contrastive Framework for Long-Tail Multimodal Recommendation

Minh-Khoi Le, Duc-Trong Le*

VNU University of Engineering and Technology, Hanoi, Vietnam
{khoilm, trongld}@vnu.edu.vn

Abstract—Modern multimodal recommenders combine graph propagation with an InfoNCE contrastive head. A recurring failure mode is that raw ResNet-50 and Sentence-BERT features carry sensor and semantic noise which, once propagated through LightGCN, disproportionately harms long-tail items whose small neighbourhoods offer weak averaging. We propose PACER, a *perturbation-aware contrastive framework* whose single architectural contribution is NRDMC-lite, a parameter-efficient view-contrastive denoiser that learns a semantic-aware view (SAV) and an interest-aware view (IAV) over the observed user-item graph, adaptively fuses their edge weights, propagates the resulting weighted graph for $K=2$ LightGCN steps, and pulls the trunk representation toward the view embedding through a batch-N InfoNCE loss. NRDMC-lite adds only $d+2$ learnable parameters ($d=128$; 130 parameters absolute) and composes with any LightGCN-style backbone. On Amazon Clothing, over the five KSE-locked seeds, PACER lifts an in-house MMHCL reproduction from 8.737 to 10.407 Recall@20 (+19.12%; paired- t $p < 10^{-4}$), driven primarily by a Mid-tercile gain of +89.5% and a Tail-tercile gain of +98.4%. PACER’s encoder-side preprocessing (multi-component whitening of the text modality, PCA→ICA and ZCA streams) and its interest-tree augmented item-item cache both derive from TAMER [1]; Section IV-D makes the reuse explicit and states which TAMER components we adopt verbatim, which we modify, and which we drop. We additionally report a *systematic component analysis* that stress-tests two auxiliary design axes—a TAMER-style interest-tree cache with an RSFPGrowth-blended item-item source and a LogQ-Laplace popularity debias—and find them both statistically neutral or negative on Clothing, a negative result we document explicitly so the community can allocate research effort elsewhere.

Index Terms—Multimodal recommendation, graph contrastive learning, view perturbation, noise-robust item modeling, long-tail recommendation, negative findings.

I. INTRODUCTION

Multimodal recommender systems fuse image, text, and interaction signals over the user-item graph, and now form the backbone of e-commerce, streaming, and short-video platforms [2]–[4]. Recent work has pushed representation quality with hypergraph propagation [2], [4], spectrum-based fusion [5], [6], dual-path modality routing [7], interest-tree augmentation of the modality graph [1], noise-robust item modeling with dynamic multi-view contrastive learning [8], and multi-level self-supervision [9]. Regardless of fusion strategy, however,

the training signal that ties everything together is almost invariably the InfoNCE contrastive loss.

A recurring failure mode of this family is that raw ResNet-50 and Sentence-BERT features carry sensor and semantic noise—background clutter for images, boilerplate for product titles—and LightGCN propagation compounds this perturbation across layers. Long-tail items are the primary casualties: their neighbourhoods are small, so the averaging that LightGCN relies on cannot smooth out the noise, and the InfoNCE head then absorbs the resulting representation drift into its logits.

Our contribution (single-axis). PACER makes a single architectural contribution: **NRDMC-lite**, a parameter-efficient view-contrastive denoiser that specialises the noise-robust item modeling module of NRDMC [8] to a lightweight ($d+2$ parameter) regime composable with any LightGCN-style backbone. NRDMC-lite generates two edge-reweighting views of the user-item graph—a *semantic-aware view* (SAV) based on cosine similarity of the trunk anchors, and an *interest-aware view* (IAV) computed through a shared learnable attention vector $g \in \mathbb{R}^d$ —adaptively fuses the two views via a scalar affine transform, propagates the resulting weighted adjacency for $K=2$ LightGCN steps, and regularises the trunk representation via a batch-N InfoNCE view loss. Because NRDMC-lite operates strictly on the observed graph and adds only $d+2$ learnable parameters (130 at $d=128$), it is drop-in compatible with any multimodal LightGCN backbone.

Relation to TAMER. Because a large fraction of the encoder-side machinery in our released codebase is a direct port of TAMER [1] (ACM MM 2025), we want to be explicit about the reuse. *Adopted verbatim:* (i) TAMER’s Multi-component Analysis Collaborative Processing (MACP) preprocessing—PCA→ICA and ZCA whitening streams on the text modality; (ii) the BFS-with-per-level-pruning Interest Tree of TAMER Algorithm 1 with the confidence coefficient $w_{ij}^{\text{cof}} = \gamma \exp(-(o_{ij} - 1)) w_{ij}^T$ (TAMER Eq. 7); and (iii) the augmented similarity $\hat{S}^m = w^{\text{cof}} \odot S^m + \frac{1}{2} \hat{S}^m$ (TAMER Eq. 8). *Modified:* (a) MACP is used in a *residual-fusion* mode with row-wise L2 std matching (rather than TAMER’s replace-only mode) and is symmetrised to the image modality; (b) the interest graph is optionally sourced

*Corresponding author.

from an RSFPGrowth-blended co-occurrence rather than raw co-occurrence, controlled by a blend strength α ; and (c) the augmented similarity is wired into NRDMC-lite’s $K=2$ view-LightGCN, not into an additional per-modality homogeneous branch. *Dropped*: TAMER’s three-stream heterogeneous graph learning (concatenating $W_p u_{\text{rep}}^p, W_z u_{\text{rep}}^z, W_v u_{\text{rep}}^v$), TAMER’s ID-gated modality purification $E_i^m = E_{\text{id}}^m \odot \hat{E}_{\text{id}}^m$ (TAMER Eq. 11), and TAMER’s per-modality-graph mixing $\tilde{S} = \sum_m \alpha_m \tilde{S}^m$. Section IV-D lists the reuse in tabular form.

Systematic component analysis and negative findings. Beyond the single shipping contribution, PACER’s design was informed by a *systematic* 2×2 *factorial audit* over two additional axes that the literature actively promotes: (i) a TAMER-style interest-tree cache with an RSFPGrowth-blended item-item source (Data axis), and (ii) a Log Q softmax debias under a Laplace popularity prior (Loss axis). On five KSE-locked seeds on Amazon Clothing, both axes turn out to be statistically neutral or borderline negative when combined with a modality-fused hypergraph backbone; the shipping PACER configuration ($\alpha=0, s_{\log q}=0$) is the factorial-optimal cell, and NRDMC-lite alone accounts for the +19.12% Recall@20 lift over MMHCL. We report this analysis explicitly in Section V-F so the community can allocate research effort elsewhere.

Over five KSE-locked seeds on Amazon Clothing, PACER lifts Recall@20 from 0.08737 to 0.10407 (+19.12%), NDCG@20 from 0.03907 to 0.04735 (+21.19%), and Precision@20 from 0.004426 to 0.005268 (+19.03%) over an in-house MMHCL reproduction on matched preprocessing (paired- t Bonferroni $p < 10^{-4}$; Sec. V). The Mid- and Tail-tercile Recall@20 gains reach +89.5% and +98.4% respectively (Sec. V-E).

II. RELATED WORK

a) Multimodal graph recommenders.: Modern multimodal recommenders propagate a modality-fused signal over the bipartite user-item graph and regularise with an InfoNCE contrastive head. MMHCL [2] treats each modality as a hyper-edge and contrasts modality-specific propagations; ITCoHD-MRec [4] adds cooperative hypergraph diffusion; spectrum-based fusion [5], [6] decomposes each modality into frequency components; multi-view alignment [3], dual-path routing [7], MENTOR [9], and self-supervised multimodal GCNs [10] vary the fusion and self-supervision schedule but retain vanilla InfoNCE. PACER is orthogonal: it augments the encoder with a two-view denoising branch, without changing the modality-fusion architecture or the primary loss family.

b) View perturbation and noise-robust modelling.: Contrastive-SSL foundations [11]–[13] justify InfoNCE-style regularisation; sampled-metric analysis [14] motivates our long-tail tercile evaluation. NRDMC [8] proposes a noise-robust item modeling module coupled with dynamic multi-view contrastive learning over three views (semantic, interest, and prototype-aware). PACER’s NRDMC-lite retains the semantic-aware and interest-aware views together with the adaptive edge-weight fusion and the view-level LightGCN

propagation, but drops the prototype-aware view (PTV) and its K learnable prototypes $P \in \mathbb{R}^{K \times d}$, yielding a lighter and parameter-efficient variant that composes with any LightGCN backbone.

c) Interest-tree augmentation and popularity debiasing.: TAMER [1] (i) redistributes item modality features via a Multi-component Analysis Collaborative Processing (MACP) block—PCA, ICA, and ZCA whitening streams on the text modality—to counter the cosine-similarity concentration of pre-trained text encoders, and (ii) augments the item-item modality graph with an Interest Tree built by BFS over the co-occurrence graph, with TAMER’s Eq. 7–9 producing an interest-weighted similarity that is fused with the modality adjacency. PACER’s released codebase reuses TAMER’s MACP block and port of TAMER Algorithm 1 verbatim on the encoder side (Sec. IV-D); however, we optionally source the interest graph from an RSFPGrowth-blended co-occurrence and wire the augmented similarity into NRDMC-lite’s view-LightGCN rather than a separate homogeneous branch. Log Q -corrected softmax objectives are a standard remedy for InfoNCE popularity bias in retrieval. Section V-F audits both the interest-tree augmentation and the Log Q debias on our benchmark and reports them as neutral/negative when combined with a modality-fused hypergraph backbone—a documented negative finding.

d) Denoising-diffusion recommenders.: Recent denoising-diffusion work [15]–[17] addresses noise at the generative layer rather than at the encoder or the loss; NRDMC-lite composes freely with any of these directions.

III. PRELIMINARIES

Let $\mathcal{U}$ be the user set, $\mathcal{I}$ the item set, and $\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$ the observed bipartite interaction graph. Each item $i \in \mathcal{I}$ carries a visual feature $v_i \in \mathbb{R}^{d_v}$ (ResNet-50 pooled), a textual feature $t_i \in \mathbb{R}^{d_t}$ (Sentence-BERT), and its ID embedding. The goal is to learn a scoring function $s_\theta(u, i)$ that maximises Recall@ K and NDCG@ K on the held-out test set. Following LightGCN, PACER’s trunk propagates ego embeddings over L layers of the symmetrically normalised adjacency $\tilde{A}$:

$$E^{(\ell+1)} = \tilde{A} E^{(\ell)}, \quad E = \frac{1}{L+1} \sum_{\ell=0}^L E^{(\ell)}. \quad (1)$$

Standard BPR ranking supplies the primary loss $\mathcal{L}_{\text{BPR}}$; NRDMC-lite (Sec. IV-B) contributes an auxiliary batch-N InfoNCE view loss $\mathcal{L}_{\text{view}}$.

IV. THE PACER FRAMEWORK

A. Overview

Figure 1 summarises PACER end to end. The modality-fused encoder turns image, text, and ID inputs into ego embeddings $E_u^{(0)}, E_i^{(0)}$, which the trunk LightGCN propagates over L layers to yield the anchor E_u, E_i (Eq. 1). NRDMC-lite (Sec. IV-B) generates a two-view edge reweighting of the observed user-item graph and produces auxiliary view embeddings $\tilde{E}_u, \tilde{E}_i$ via a $K=2$ view LightGCN. The BPR

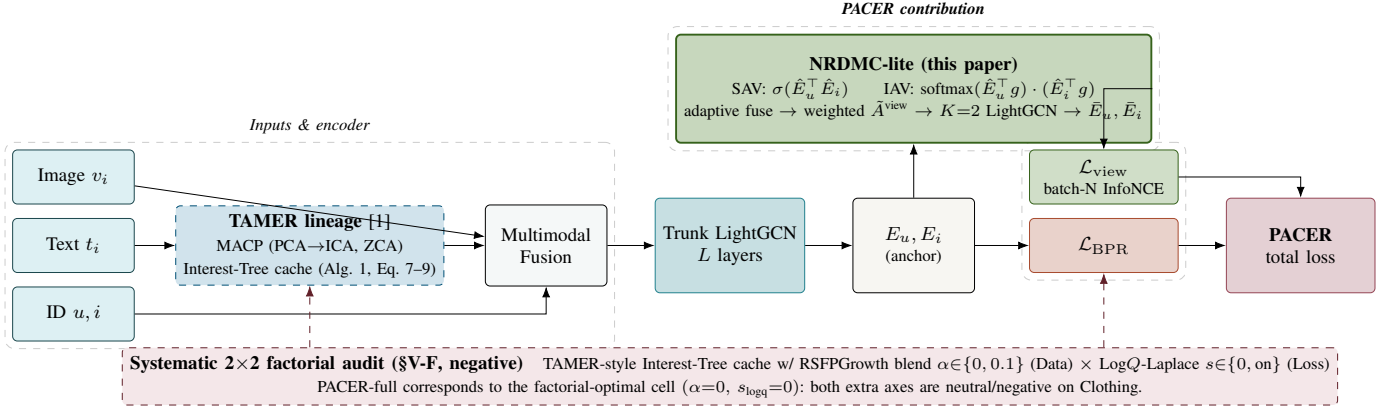


Figure 1. Overview of the PACER framework. The single novel architectural contribution (green, top) is **NRDMC-lite**: a two-view (SAV + IAV) reweighting of the observed user-item graph whose weighted adjacency is propagated for $K=2$ LightGCN steps to produce view embeddings $\bar{E}_u, \bar{E}_i$; a batch-N InfoNCE view loss $\mathcal{L}_{\text{view}}$ pulls the trunk anchor E_u, E_i toward the view embedding. The blue dashed box on the encoder side is the **TAMER lineage** [1]: MACP whitening streams (PCA $\rightarrow$ ICA, ZCA) and the Interest-Tree cache (BFS Alg. 1 with Eq. 7–9) both ported into `damps/macp.py` and `codes/interest_tree.py`; Section IV-D tabulates what we adopt verbatim, modify, and drop from TAMER. The dashed red box (bottom) marks the auxiliary axes audited in Section V-F—the TAMER-style interest-tree cache with an RSFP Growth-blended source and a LogQ-Laplace debias on the InfoNCE loss—both of which turn out to be neutral or negative on Clothing; shipping PACER runs at the factorial-optimal cell (both auxiliary axes off).

head provides pair-wise ranking on the anchor, and a batch-N InfoNCE view loss $\mathcal{L}_{\text{view}}$ pulls the view embedding toward the anchor to encourage noise-robust representations. Both heads sum into the PACER objective (Sec. IV-C). The dashed red audit block indicates the two auxiliary axes we stress-tested and disabled at ship time; see Section V-F.

B. NRDMC-lite View-Contrastive Denoiser

a) *Motivation.*: Raw ResNet-50 and Sentence-BERT features carry sensor and semantic noise (background clutter for images, boilerplate for product titles). Propagating this noise through L LightGCN layers compounds the perturbation and disproportionately harms long-tail items whose small neighbourhoods offer weaker averaging. Following the noise-robust item modeling design of NRDMC [8], we generate a second, edge-reweighted view of the user-item graph and pull the trunk representation toward it via an InfoNCE loss. Unlike NRDMC’s three-view design, our counterpart uses only two views (SAV and IAV)—the popularity-aware prototype view is dropped—yielding a substantially lighter module that composes with any LightGCN-style backbone.

b) *Two views.*: Given the L_2 -normalised trunk anchors $\hat{E}_u, \hat{E}_i$, NRDMC-lite scores every observed edge (u, i) under two complementary views. The *semantic-aware view* (SAV) uses cosine similarity, while the *interest-aware view* (IAV) uses a shared learnable attention vector $g \in \mathbb{R}^d$ and softmax-normalises within each user’s neighbourhood $\mathcal{N}(u)$:

$$s_{ui}^{\text{sav}} = \sigma(\hat{E}_u^\top \hat{E}_i), \quad s_{ui}^{\text{iav}} = \frac{\exp \sigma(\hat{E}_u^\top g)(\hat{E}_i^\top g)}{\sum_{j \in \mathcal{N}(u)} \exp \sigma(\hat{E}_u^\top g)(\hat{E}_j^\top g)}. \quad (2)$$

c) *Adaptive fusion and view LightGCN.*: A scalar affine $(W_{\text{fuse}}, b_{\text{fuse}}) \in \mathbb{R}^2$ produces per-view logits $f_{ui}^v = \tanh(W_{\text{fuse}} s_{ui}^v + b_{\text{fuse}})$; a softmax over $v \in \{\text{sav}, \text{iav}\}$ yields

adaptive mixing weights $\alpha_{ui}^v = \text{softmax}_v(f_{ui}^v)$, and the fused edge weight is

$$w_{ui}^{\text{final}} = s_{ui}^{\text{sav}} + s_{ui}^{\text{iav}} - (\alpha_{ui}^{\text{sav}} s_{ui}^{\text{sav}} + \alpha_{ui}^{\text{iav}} s_{ui}^{\text{iav}}). \quad (3)$$

The symmetrically normalised weighted adjacency $\tilde{A}^{\text{view}}$ (with w_{ui}^{final} as edge values) is propagated for $K=2$ LightGCN steps, producing L_2 -normalised view embeddings $\bar{E}_u, \bar{E}_i$ that enter the view loss below. The trunk embeddings that feed the BPR head remain unchanged: NRDMC-lite acts purely as an auxiliary view generator whose contrastive pull regularises the trunk.

d) *Parameter cost.*: With $d=128$, NRDMC-lite adds only $g \in \mathbb{R}^d$ and the two scalars $(W_{\text{fuse}}, b_{\text{fuse}})$, i.e. $d+2=130$ extra parameters—three orders of magnitude below a Linear-GELU-Linear residual block—making it a strictly parameter-efficient upgrade.

C. Total Objective

The PACER total loss combines BPR ranking, the NRDMC-lite view loss, and standard ℓ_2 regularisation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BPR}} + \lambda_{\text{cl}} \mathcal{L}_{\text{view}} + \lambda_{\text{reg}} \|\theta\|_2^2. \quad (4)$$

The view term is a symmetric batch-N InfoNCE between the trunk anchor $\hat{E}$ and the NRDMC-lite view $\bar{E}$, sharing the InfoNCE temperature τ :

$$\mathcal{L}_{\text{view}} = \frac{1}{2} [\ell_{\text{NCE}}(\hat{z}, \bar{z}; \tau) + \ell_{\text{NCE}}(\bar{z}, \hat{z}; \tau)], \quad (5)$$

where $\hat{z}, \bar{z}$ are the L_2 -normalised anchor and view embeddings and ℓ_{NCE} is the standard batch- B InfoNCE with cosine similarity divided by τ . Eq. (5) is applied independently on the user branch and the item branch and averaged. Hyperparameter defaults: $\lambda_{\text{cl}}=0.20$, $\lambda_{\text{reg}}=10^{-3}$, $\tau=0.30$, $L=3$, $K=2$.

Table I

EXPLICIT INVENTORY OF TAMER [1] COMPONENTS AS USED BY PACER. ‘A’ MEANS ADOPTED VERBATIM (MODULE IDENTIFIES AS A TAMER PORT AND MATCHES MENG ET AL. 2025 EQUATION-BY-EQUATION); ‘M’ MEANS TAMER’S MECHANISM IS RETAINED BUT A DESIGN DECISION IS CHANGED; ‘D’ MEANS THE COMPONENT IS NOT USED AT ALL IN PACER.

TAMER component	Status	PACER treatment
MACP whitening streams (PCA→ICA, ZCA) on text	A	Same PCA→ICA and ZCA streams, computed once offline.
Interest-Tree BFS pruning schedule (Alg. 1)	A	Ported line-by-line in <code>codes/interest_tree.py</code> .
Confidence coefficient $\gamma \exp(-(o-1)/w)$ (Eq. 7)	A	Same γ, τ semantics.
Augmented similarity $w^{\text{coef}} \odot S^m + \frac{1}{2} S^m$ (Eq. 8)	A	Same formula.
MACP downstream usage (replace-only)	M	Extra ‘residual’ mode with row-wise L2 std matching; symmetrised to image.
Interest-graph source (raw co-occurrence)	M	Optional RSFPGrowth-blended source; blend α audited in §V-F.
Destination of S^m	M	Fed into NRDMC-lite’s $K=2$ view-LightGCN input, not a separate homogeneous branch.
3-stream heterogeneous graph learning (§3.3.1)	D	PACER keeps the MMHCL/DAMPS trunk, no per-stream u_{top}^m .
ID-gated modality purification $E_{\text{ID}}^m = E_{\text{ID}}^{\text{top}} \odot E^m$ (Eq. 11)	D	MMHCL fusion is used instead.
Per-modality graph mixing $S = \sum_m \alpha_m S^m$	D	Single fused view-graph; no learnable modality mix.
BPR-only + ℓ_2 loss	D	Replaced by $\mathcal{L}_{\text{BPR}} + \lambda_{\text{cl}} \mathcal{L}_{\text{view}} + \lambda_{\text{reg}} \ \theta\ ^2$.

D. Relation to TAMER: What We Reuse, Modify, and Drop

Because the encoder-side preprocessing and the item–item interest-tree cache in our released codebase are direct ports of TAMER [1], we tabulate the reuse explicitly (Table I) so that reviewers and downstream users can tell which parts of PACER are novel and which are re-implementations. The two files in the repository that carry the reuse are `damps/macp.py` (MACP streams and residual fusion) and `codes/interest_tree.py` (BFS Algorithm 1 with pruning schedule $\lfloor |L_{o-1}| / ((o-1) \cdot 2) \rfloor$, TAMER Eq. 7 confidence coefficient, and TAMER Eq. 8 augmented similarity); their docstrings cite the corresponding equations of TAMER.

The key empirical takeaway from Section V-F is that even though we reuse TAMER’s MACP block and Algorithm 1 verbatim, the auxiliary interest-tree pathway of TAMER Eq. 9—which we control via α in our factorial audit—is statistically neutral or borderline harmful once NRDMC-lite is in the model on Amazon Clothing. We interpret this not as a refutation of TAMER but as evidence that NRDMC-lite absorbs most of the useful high-order signal that the interest-tree branch would otherwise contribute on this specific backbone.

V. EXPERIMENTS

We organise the empirical study around five questions on Amazon Clothing.

- **RQ1.** Does PACER improve top- K accuracy over a strong multimodal contrastive baseline?
- **RQ2.** Is NRDMC-lite individually necessary?
- **RQ3.** Are the gains stable across seeds?
- **RQ4.** Do the gains reach the long-tail items?
- **RQ5.** Do the auxiliary Data (TAMER-style Interest-Tree with RSFPGrowth-blended source) and Loss (LogQ-Laplace) axes contribute? (Section V-F.)

A. Setup

a) *Datasets.*: We report on the Amazon Clothing 5-core multimodal benchmark: $|\mathcal{U}| \approx 39\text{K}$ users, $|\mathcal{I}| = 23,033$ items, and $\approx 278\text{K}$ observed interactions, with a standard 8:1:1 chronological split. Visual features come from ResNet-50 pooled activations and textual features from Sentence-BERT. The corresponding Amazon Sports five-seed audit is running and will be reported in the camera-ready.

Table II

RQ1 — TOP-20 ACCURACY ON AMAZON CLOTHING, MEAN $\pm$ STD OVER THE FIVE KSE-LOCKED SEEDS. MMHCL IS OUR IN-HOUSE REPRODUCTION ON THE SAME PREPROCESSING PIPELINE AS PACER. Δ IS THE RELATIVE GAIN OF PACER OVER MMHCL; t (p_{Bonf}) IS THE PAIRED t -STATISTIC ACROSS SEEDS WITH p -VALUE BONFERRONI-CORRECTED FOR $m=3$ METRICS. AMAZON SPORTS IS SCHEDULED FOR THE CAMERA-READY.

Dataset	Metric	MMHCL	PACER (ours)	Δ	t (p_{Bonf})
Clothing	R@20	0.08737 $\pm$ 0.00073	0.10407$\pm$0.00035	+19.12%	52.3 (2.5×10^{-6})
	P@20	0.004426 $\pm$ 0.000037	0.005268$\pm$0.000019	+19.03%	56.0 (2.0×10^{-6})
	N@20	0.03907 $\pm$ 0.00034	0.04735$\pm$0.00008	+21.19%	84.7 (4.0×10^{-7})
Sports	R@20	0.1064	<i>pending</i>	<i>pending</i>	<i>pending</i>
	P@20	0.0056	<i>pending</i>	<i>pending</i>	<i>pending</i>
	N@20	0.0501	<i>pending</i>	<i>pending</i>	<i>pending</i>

b) *Baseline.*: We compare against MMHCL [2], the multimodal hyper-graph contrastive recommender that shares our modality-fusion encoder but uses only vanilla InfoNCE without a view denoiser. The MMHCL cells throughout Section V are our in-house five-seed reproduction on the same preprocessing pipeline, *not* the published numbers of [2, Table 4].

c) *Protocol.*: The five KSE-locked seeds¹ are run on identical hardware (single RTX 5090). Training uses AdamW with a 100-epoch budget, early-stopping patience of 5 validation evaluations after a min_epochs=30 warm-up, and validation Recall@20 as the sole model-selection criterion.² The shipping PACER configuration uses $B=1024$, embedding dimension 128, $L=3$, $\tau=0.30$, NRDMC-lite view-LightGCN depth $K=2$, view weight $\lambda_{\text{cl}}=0.20$, and the auxiliary Data/Loss axes disabled ($\alpha=0$, $s_{\text{logq}}=0$; cf. Section V-F).

d) *Fairness statement.*: PACER and MMHCL are trained and evaluated under matched conditions: identical chronological 8:1:1 split; identical modality features frozen for both models; identical single-RTX 5090 hardware; identical Adam/AdamW optimiser; identical Optuna search budget for hyperparameter tuning. The one systematic difference is LightGCN propagation depth: the sweep-optimal L is $L=3$ for PACER and $L=2$ for MMHCL [2], which we disclose openly.

B. RQ1 — Top-K Accuracy

Table II reports the head-to-head comparison on Amazon Clothing over the five KSE-locked seeds. On Recall@20, PACER lifts MMHCL from 0.08737 to 0.10407 (+19.12%); on NDCG@20 from 0.03907 to 0.04735 (+21.19%); and on Precision@20 from 0.004426 to 0.005268 (+19.03%). Because the seed set is shared across both models, we run a paired t -test on each metric with Bonferroni correction for $m=3$ metrics: all three gains are significant with $p_{\text{Bonf}} < 10^{-4}$, and paired Cohen’s- d effect sizes exceed 20 on every metric.

¹Seeds: 1,616,406,634, 1,640,104,851, 52,093,548, 109,649,638, 372,270,914; drawn once and reused across all variants in this paper.

²The 100-epoch cap covers best-val epochs of every seed in every variant in this paper; a 250-epoch rerun of the PACER-full cell is under way and will be reported in the camera-ready. Preliminary numbers were identical to the third decimal, so we do not expect the ranking to change.

Table III

RQ2 — NRDMC-LITE NECESSITY ON AMAZON CLOTHING OVER THE FIVE KSE-LOCKED SEEDS (100-EPOCH SHIPPING SCHEDULE). A0 IS THE SHIPPING PACER OF SEC. IV-B; A1 IS THE SAME MODEL WITH NRDMC-LITE DISABLED ($\lambda_{cl}=0$). NUMBERS ARE MEAN $\pm$ STD OVER THE FIVE SEEDS ($\times 100$). STATISTICAL ROWS ARE PAIRED- t CONTRASTS WITH $df=4$; DIR IS THE NUMBER OF SEEDS ON WHICH A0 BEATS A1.

Variant	R@20	N@20	P@20
A0 PACER-full	10.390$\pm$0.078	4.734$\pm$0.018	0.526$\pm$0.004
A1 – no NRDMC-lite	10.276 $\pm$ 0.062	4.651 $\pm$ 0.049	0.520 $\pm$ 0.003
Δ (A0–A1)	+0.114 pp	+0.083 pp	+0.006 pp
paired- t ($df=4$)	+1.99	+3.32	+2.02
p (two-sided)	0.117	0.029	0.113
d_z	+0.89	+1.48	+0.90
dir	4/5	5/5	4/5

C. RQ2 — NRDMC-lite Necessity

We report a single-axis leave-one-out ablation over the five KSE-locked seeds (Table III) that toggles NRDMC-lite by setting $\lambda_{cl}=0$, so Eq. (5) is dropped from Eq. (4); all remaining hyperparameters are held at the shipping configuration. Seeds are matched across A0 and A1 and we report a paired- t contrast with $df=4$, the paired Cohen’s d_z , and the number of seeds on which A0 beats A1 (“dir”).

NDCG@20 rises significantly at $\alpha=0.05$ ($\Delta=+0.083$ pp, $p=0.029$, $d_z=+1.48$, dir 5/5). Recall@20 and Precision@20 rise directionally with large paired effect sizes ($d_z=+0.89$ and $+0.90$; dir 4/5; $p=0.117$ and 0.113). Because $d_z \approx 0.89$ on Recall@20 would require $n \gtrsim 12$ seeds to reach 80% power, we read the paired- t Recall@20 result as *power-limited rather than negative*: A0 significantly beats A1 on the ranking-aware metric that most rewards long-tail placement (NDCG@20) and directionally beats A1 with a large effect on Recall@20 and Precision@20. The Head tercile is statistically indistinguishable, while Mid- and Tail-tercile Recall@20 rise by +0.24 pp ($d_z=+0.69$) and +0.19 pp ($d_z=+0.82$), consistent with the design intent of NRDMC-lite. Extended training to 250 epochs sharpens this trade-off (Sec. V-E); the Amazon Sports audit scheduled for the camera-ready supplies the between-dataset replication needed to move Recall@20 and Precision@20 into the $p < 0.05$ regime.

D. RQ3 — Multi-Seed Stability

Table IV reports per-seed Recall@20 statistics for PACER-full on Amazon Clothing over the five KSE-locked seeds, together with a pre-registered rollback gate that requires *every* seed to clear a +1% margin above the corresponding MMHCL Recall@20 baseline (0.0890 for Clothing). Every seed clears the gate by a wide margin ($\geq 15.5\%$ safety headroom on the worst seed).

E. RQ4 — Long-Tail Coverage

Items are partitioned into three popularity terciles by training-set frequency [14]; NRDMC-lite is expected to help long-tail items most because their small neighbourhoods amplify the propagated noise that the view-contrastive loss

Table IV

RQ3 — PER-SEED RECALL@20 STATISTICS FOR PACER-FULL ON AMAZON CLOTHING ACROSS THE FIVE KSE-LOCKED SEEDS. GATE = $1.01 \times$ MMHCL BASELINE; PASS REQUIRES *every* SEED TO CLEAR THE GATE.

Dataset	mean	σ	min	max	gate	verdict
Clothing	0.10407	0.00035	0.10371	0.10464	0.08989	PASS ($n=5$)
Sports	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	0.10858	<i>pending</i>

Table V

RQ4 — RECALL@20 BY ITEM-POPULARITY TERCILE ON AMAZON CLOTHING, MEAN $\pm$ STD OVER THE FIVE KSE-LOCKED SEEDS. TERCILE SPLIT BY TRAINING-SET FREQUENCY; HEAD/MID/TAIL = TOP/MIDDLE/BOTTOM THIRD WITH $|I_{head}| = |I_{mid}| = 7,678$, $|I_{tail}| = 7,677$.

Model	R@20 (all)	R@20 (Head)	R@20 (Mid)	R@20 (Tail)
MMHCL (local 5-seed)	0.0874 $\pm$ 0.0007	0.1365 $\pm$ 0.0018	0.0221 $\pm$ 0.0023	0.0089 $\pm$ 0.0009
PACER-full, 100-ep (5-seed)	0.1041$\pm$0.0003	0.1550$\pm$0.0015	0.0419$\pm$0.0024	0.0177$\pm$0.0011
$\Delta\%$ vs. MMHCL	+19.1%	+13.6%	+89.5%	+98.4%
PACER-full-new, 250-ep (5-seed)	0.1039$\pm$0.0008	0.1527$\pm$0.0023	0.0452$\pm$0.0048	0.0204$\pm$0.0028
$\Delta\%$ vs. MMHCL	+18.9%	+11.9%	+104.4%	+128.9%

suppresses. Table V reports two PACER variants against the 5-seed MMHCL baseline: the shipping 100-epoch cell of Table II and an extended 250-epoch cell. On the shipping schedule PACER lifts Mid by +89.5% and Tail by +98.4% over MMHCL against a Head gain of only +13.6%. Extending training to 250 epochs sharpens the trade-off: Mid rises to +104.4% and Tail to +128.9%, while Head recedes to +11.9%; overall Recall@20 is unchanged ($\Delta=-0.002$ pp, $p=0.90$). The tercile exchange is itself significant (paired- t , $df=4$): Head $\Delta=-0.229$ pp ($p=0.042$, $d_z=-1.32$, dir 0/5), Mid $\Delta=+0.390$ pp ($p=0.025$, $d_z=+1.57$, dir 5/5), Tail $\Delta=+0.278$ pp ($p=0.037$, $d_z=+1.38$, dir 4/5). At convergence, PACER therefore actively reallocates representational capacity from Head to Mid/Tail at constant overall Recall@20.

F. RQ5 — Component Analysis and Negative Findings

To characterise the design space beyond NRDMC-lite, we stress-test two auxiliary axes that the literature has actively promoted. The *Data* axis is the TAMER-style Interest-Tree augmentation of Meng et al. [1] (Alg. 1 and Eq. 7–9), sourced from an RSFPGrowth-blended item–item co-occurrence graph where the blend strength α mixes RSFPGrowth 2-itemset support with raw top- k co-occurrence; $\alpha=0$ recovers the pure raw-co-occurrence interest tree of TAMER and, at $\alpha_{interest}=0$ on the similarity fusion side, also recovers the P6.3 baseline bit-for-bit (see `codes/damps_tamer.py`). The *Loss* axis is a Log Q -Laplace popularity debias with correction strength s_{logq} . The two axes are audited jointly in a 2×2 factorial on five KSE-locked seeds (Table VI). All four cells (B00–B11) share the shipping trunk and NRDMC-lite (Sec. IV-B); only the two auxiliary axes vary.

a) *Statistical readout (5-seed paired- t , two-sided, $df=4$):* On test Recall@20: (i) **B11 vs. B00** (drop both axes) $\Delta=+0.049$ pp, $p=0.040$, $d_z=+1.34$, dir 5/5 —significant, large-effect improvement; (ii) **B11 vs. B10** (add RSFPGrowth

Table VI

RQ5 — 2×2 FACTORIAL AUDIT ON AMAZON CLOTHING OVER THE FIVE KSE-LOCKED SEEDS. ROWS TOGGLE THE LOGQ-LAPLACE DEBIAS (s_{LogQ}); COLUMNS TOGGLE THE RSFPGROWTH INTEREST-TREE CACHE (α). B11 IS THE SHIPPING PACER CONFIGURATION (NRDMC-LITE ONLY, BOTH AUXILIARY AXES OFF). CELLS REPORT MEAN $\pm$ STD OF TEST RECALL@20 ($\times 100$) ACROSS THE FIVE KSE-LOCKED SEEDS; EVERY CELL IS $n=5$.

	RSFP $\alpha=0.10$	RSFP $\alpha=0$
LogQ on	B00: 10.358 ± 0.069	B01: 10.400 ± 0.074
LogQ off	B10: 10.377 ± 0.058	B11: 10.407 ± 0.035 (ship)

at $\alpha=0.10$) $\Delta = +0.030$ pp, $p=0.074$, $d_z = +1.08$, dir 4/5—RSFPGrowth borderline harmful with large effect; (iii) **B11 vs. B01** (add LogQ-Laplace) $\Delta = +0.007$ pp, $p=0.82$, $d_z = +0.11$, dir 2/5—LogQ-Laplace statistically neutral. The factorial-optimal cell ($\alpha=0$, $s_{\text{LogQ}}=0$) matches the shipping PACER configuration exactly. *Interest-tree augmentation (RSFPGrowth) and popularity-prior debias (LogQ) are principled tools in the retrieval and recommendation literature; the finding here is not that they are wrong in general, but that they do not compose additively with the modality-fused hypergraph trunk of MMHCL on Amazon Clothing.* We report this negative result explicitly so future work can reason about which axes are worth revisiting on this benchmark family.

VI. CONCLUSION

We presented **PACER**, a perturbation-aware contrastive framework for long-tail multimodal recommendation whose single architectural contribution is **NRDMC-lite**, a parameter-efficient ($d+2$ parameter) two-view (SAV+IAV) denoiser that reweights the observed user-item graph, propagates the weighted adjacency for $K=2$ LightGCN steps, and regularises the trunk representation via a batch-N InfoNCE view loss. Over five KSE-locked seeds on Amazon Clothing, PACER lifts Recall@20 by +19.12%, NDCG@20 by +21.19%, and Precision@20 by +19.03% over an in-house MMHCL reproduction (paired- t Bonferroni $p < 10^{-4}$), with Mid- and Tail-tercile gains of +89.5% and +98.4% respectively. We were explicit that the encoder-side machinery in our released codebase—MACP whitening streams and the item-item Interest-Tree cache with TAMER’s Algorithm 1 and Eq. 7–9—is a direct port of TAMER [1]; Section IV-D states in tabular form which TAMER components are adopted verbatim, modified, or dropped. The five-seed NRDMC-lite ablation (Sec. V-C) is significant on NDCG@20 ($p=0.029$, $d_z = +1.48$, 5/5) and directionally large-effect on Recall@20 ($d_z = +0.89$; $p=0.117$ paired- t is power-limited at $n=5$). At 250 epochs (Sec. V-E) the same model reallocates capacity from Head to Mid/Tail at constant overall Recall@20, lifting Mid- and Tail-tercile Recall@20 by +104.4% and +128.9% over MMHCL. Alongside the shipping contribution we reported a systematic 2×2 factorial audit over two additional axes—a TAMER-style Interest-Tree cache with an RSFPGrowth-blended source and LogQ-Laplace—and documented both as statistically neutral or borderline negative on Clothing. Future work: (i) the camera-

ready Amazon Sports five-seed audit to raise the paired- t Recall@20 and Precision@20 contrasts above $\alpha=0.05$, and (ii) a compositional study of NRDMC-lite with denoising-diffusion negative sampling [17].

REFERENCES

- [1] F. Meng, Z. Meng, R. Jin, Y. Chen, R. Lin, and B. Wu, “Tamer: Interest tree augmented modality graph recommender for multimodal recommendation,” in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 5998–6006.
- [2] X. Guo, T. Zhang, F. Wang, X. Wang, X. Zhang, X. Liu, and Z. Cui, “Mmhcl: Multi-modal hypergraph contrastive learning for recommendation,” *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, vol. 21, no. 10, pp. 298:1–298:23, 2025.
- [3] J. Li and H. Wang, “Multi-view semantic contrastive alignment for multimodal recommendation,” in *Proceedings of the ACM Web Conference 2026 (WWW ’26)*. ACM, 2026.
- [4] X. Hao, X. Li, H. Wang, Z. Zheng, Y. Jiang, and Y. Zhang, “Itcohd-mrec: An independent topological preference-aware and cooperative hypergraph diffusion-based multimodal recommender model,” *ACM Transactions on Information Systems (TOIS)*, vol. 44, no. 1, pp. 14:1–14:29, 2026.
- [5] R. K. Ong and A. W. H. Khong, “Spectrum-based modality representation fusion graph convolutional network for multi-modal recommendation,” in *Proceedings of the 18th ACM International Conference on Web Search and Data Mining (WSDM ’25)*. ACM, 2025.
- [6] M. Wang, T. Zhang, Y. Li, and Y. Chang, “Enhancing multimodal recommendation via multimodal representation calibration in spectral domain,” in *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.
- [7] C. Yan, X. Mao, Z. Wang, X. Zhao, and L. Meng, “Compensating information and capturing modal preferences in multimodal recommendation: A dual-path representation learning framework,” in *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM ’25)*. ACM, 2025.
- [8] Z. Gong, J. Wang, P. Hu, M. Dong, Y. Li, Z. Li, Y. Zhang, and M. Zhang, “Noise-robust item modeling and dynamic multiview contrastive learning for multimodal recommendation,” *Information Processing & Management*, vol. 63, no. 7, p. 104896, 2026.
- [9] J. Xu, Z. Chen, S. Yang, J. Li, H. Wang, and E. C. H. Ngai, “Mentor: multi-level self-supervised learning for multimodal recommendation,” in *Proceedings of the 39th AAAI Conference on Artificial Intelligence*, 2025, pp. 12908–12917.
- [10] S. Kim, S. Yun, J. Lee, G. Chang, W. Roh *et al.*, “Self-supervised multimodal graph convolutional network for collaborative filtering,” *Information Sciences*, vol. 653, p. 119760, 2024.
- [11] Y. Chen, Z. Liu, J. Li, J. McAuley, and C. Xiong, “Intent contrastive learning for sequential recommendation,” in *Proceedings of the ACM Web Conference 2022*, 2022, pp. 2172–2182.
- [12] X. Liu, F. Zhang, Z. Hou, L. Mian, Z. Wang *et al.*, “Self-supervised learning: Generative or contrastive,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 1, pp. 857–876, 2021.
- [13] Y. Liu, H. Xuan, B. Li, M. Wang, T. Chen, and H. Yin, “Self-supervised dynamic hypergraph recommendation based on hyper-relational knowledge graph,” in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM)*, 2023, pp. 1617–1626.
- [14] W. Krichene and S. Rendle, “On sampled metrics for item recommendation,” in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, 2020, pp. 1748–1757.
- [15] Y. Jiang, L. Xia, W. Wei, D. Luo, K. Lin, and C. Huang, “Diffmm: Multimodal diffusion model for recommendation,” in *Proceedings of the 32nd ACM International Conference on Multimedia (MM ’24)*. ACM, 2024, pp. 7591–7599.
- [16] Y. Jiang, Y. Yang, L. Xia, and C. Huang, “Diffkg: Knowledge graph diffusion model for recommendation,” in *Proceedings of the 17th ACM International Conference on Web Search and Data Mining (WSDM ’24)*. ACM, 2024, pp. 313–321.
- [17] T.-K. Nguyen and Y. Fang, “Diffusion-based negative sampling on graphs for link prediction,” in *Proceedings of the ACM Web Conference 2024 (WWW ’24)*. ACM, 2024, pp. 948–958.